

COGS 566 - Probabilistic Models of Cognition

Project Proposal

*Exploring Vagueness and Context-Sensitivity in Gradable Adjectives (Positive-Form)
Using a Probabilistic Pragmatic Model (RSA).*

Turgay Yıldız

Department of Cognitive Sciences , Middle East Technical University

Abstract :

This project aims to investigate *vagueness* and *context sensitivity* in gradable adjectives, focusing specifically on their *positive form*, through the Rational Speech Act (RSA) model. Gradable adjectives exhibit properties such as borderline cases, context-dependent thresholds, and susceptibility to phenomena like the Sorites paradox. This study will develop a computational framework to analyze the context-sensitive interpretation of gradable adjectives. Synthetic datasets inspired by existing experimental paradigms will be used to explore threshold variability and borderline cases under different comparison classes. The study will exclude intensifiers and negation to concentrate on vagueness and context sensitivity in gradable adjectives. By enhancing our understanding of these phenomena, this study seeks to contribute to the growing body of research on probabilistic pragmatics and its application to linguistic semantics.

A. Literature Survey

1. Key Concepts :

Adjectives can be categorized into scalar (gradable) and non-scalar (non-gradable) types. Scalar adjectives are further divided into *absolute* and *relative* types. Absolute adjectives can also be classified as total or partial (Burnett, 2014).

Gradable adjectives can be used in superlative and comparative constructions and can be modified by intensifiers (Zhao., 2019). For example, Zhao (2019) gives the word “tall” as an example: it can be used as “taller” (comparative construction), as “tallest” (superlative constructions), and it can also be modified by intensifiers as “very tall”. Gradable adjectives are most often studied in their positive form, which refers to the morphologically simple and unmarked form. For example, “tall” and “late” are gradable adjectives in the positive form. On the other hand, adjectives like “atomic”,

“pregnant”, and “geographical” are not gradable (Burnett., 2014). This is because they cannot be used in constructions such as 'This planet is more atomic than the other one.'

Literature mostly explored the difference between absolute and relative adjectives in the positive form, how they can be differentiated by employing gradable property (degree scales) (Xiang et al., 2022; Qing and Franke., 2014; Burnett., 2014) , and how and why vagueness and context-sensitivity arise in those adjectives (Burnett., 2014; Kennedy., 2007; Lassiter and Goodman., 2013).

These studies primarily address two questions: first, what is the relationship between scale structure and the absolute/relative distinction? Second, how does vagueness relate to contextual variability, borderline cases, and the Sorites paradox? Exploration of gradable adjectives by employing their scale structure is called degree semantics. In degree semantics, meaning of the positive form contains a free variable (Lassiter and Goodman., 2013). According to degree semantics, each gradable adjective is like a function which takes an instance as input and gives a value according to its scale. For example, “tall” is a gradable adjective which can be written as

$\lambda x. \text{height}(x) \geq \theta(c)$ (Zhao., 2019). $\theta(c)$ is called threshold which depended on the context. The threshold is sometimes called standard of comparison (Qing and Franke., 2014). The fact that threshold is context-sensitive leads us to another important concept called comparison classes.

Comparison classes are crucial because, when they change, the standard of comparison (threshold) for relative adjectives also changes. Eventually, this bring the answer to the first question: what is the relationship between scale structure and absolute/relative distinction. Before elaborating this answer, we first need to mention some other key phenomenon and relationships between them. Those phenomenon are borderline cases, shifting the threshold, open and closed scales, Sorites Paradox, and tolerance. Borderline cases, and shifting the threshold - via comparison classes - help us to discriminate relative adjectives from absolute adjectives. Moreover, open and closed scales help us to understand the Sorites Paradox via tolerance.

Absolute adjectives like *dry*, and *straight* are associated with scales that have a *maximal* element. On the other hand, absolute adjectives like *slightly*, and *a little* are associated with scales that have a *minimal* element (Burnett., 2014). The first type of scale is called a top-closed scale, while the second type is referred to as a bottom-closed scale. Relative adjectives have open scales. Therefore, they lead to *Sorites Paradox*.

Ultimately, scale structure—due to the context-dependent threshold and the presence of borderline cases leading to hesitancy—helps distinguish relative adjectives from absolute adjectives. For instance, we can classify 'tall' as a relative adjective and 'empty' as an absolute adjective. A detailed exploration of these concepts is available in the works of Xiang et al. (2022), Burnett (2011), and Kennedy (2007).

This distinction between absolute and relative adjectives will be explored in the context of vagueness and context-sensitivity in this study.

2. Models :

Burnett (2014) compares the delineation semantics framework (DeIS) with the degree semantics framework. He proposes a new logical system by integrating the Tolerant, Classical, Strict (TCS) framework with DeIS. He calls this new framework Delineation TCS (DeITCS), through which he analyses absolute scalar adjectives (AAs), relative scalar adjectives (RAs), and non-scalar adjectives (NSs). He begins with a basic model called Classical Model (C-Model). This model interprets predicates according to comparison classes.

Lassiter and Goodman (2013) use a pragmatic model where speakers and listeners jointly infer the state of the world and semantic variable values to determine posterior distributions. They refer to this cooperation between listeners and speakers as coordination. The speaker chooses utterances based on how the listener will interpret them. The listener uses models of the speaker to resolve context sensitivity via Bayes' Rule. This model is a general Rational Speech Act (RSA) framework consisting of a Literal Listener (L0), which responds to utterances by considering only their truth values; a Speaker (S1), who maximizes the probability that L0 selects the correct interpretation; and a Pragmatic Listener (L1), who infers the speaker's intent using Bayesian reasoning. The RSA model primarily focuses on predicting the threshold.

Qing and Franke (2014) critique the RSA model by Lassiter and Goodman (2013) for being listener-oriented. They propose a new model inspired by the RSA framework. The proposed model is a speaker-oriented model (SOM). They integrate the concept of optimal linguistic conventions, drawn from evolutionary approaches, into this new model.

Their primary goals in developing this model are twofold. First, they aim to explain how the standard of comparison (threshold) is contextually determined. Second, they seek to explain why gradable adjectives are distinguished as absolute or relative. They argue that in the RSA model, the speaker knows the exact threshold for the positive form. Moreover, the speaker will either utter the positive form or remain silent. Even if the speaker does not know the exact threshold, it will use positive form with equal likelihood no matter what the actual degree is.

Zhao (2019) presents two probabilistic pragmatic models: the RSA model by Lassiter and Goodman (2013) and the SOM by Qing and Franke (2014). He elaborates on both models, comparing their similarities and differences. However, unlike Lassiter and Goodman (2013) and Qing and Franke (2014), who focus on gradable adjectives in their RSA and SOM models, Zhao examines intensifiers. He references Bennett and Goodman (2018), who claim that intensifiers cause a shift in the threshold in a non-compositional way. This means they assume that different intensifiers do not alter semantics but only shift the threshold. The only difference lies in how each intensifier shifts the threshold. To account for this shift, an additional cost function is incorporated into the model.

Xiang et al. (2022) employ the RSA model by Lassiter and Goodman (2013) and the SOM model by Qing and Franke (2014). The RSA framework models language communication as a recursive process. They incorporate three variables: threshold, length-based cost, and lambda for the degree of rationality. As noted earlier, the SOM is speaker-oriented, unlike the RSA, which is listener-oriented.

3. Data, Observational and Experimental Studies :

Zhao (2019) used experimental data from Leffel et al. (2018). In this experiment, participants were asked to adjust a slider on a scale from 0 to 100 to indicate their agreement with sentences like 'Ronald is tall,' given Ronald's exact height. They also tested intensifiers, such as 'very,' along with other constructions. Thirteen different height values were used.

Xiang et al. (2022) also used experimental data from human participants. They presented subjects with items from two different categories. Although these items belonged to different categories, they were quite similar. The explicit difference was a specific scalar property, such as degree of length, height, or bend. The first category consisted of artifacts, which are objects commonly encountered in daily life, such as candles, pillows, and nails. The second category was called 'shapes,' which included abstract forms rarely encountered in daily life. They then asked participants to choose a single object from each category that they believed to be 'the most likely.' They expected that prior beliefs for the first category (artifacts) would be fine-grained, while prior beliefs for the second category (shapes) would be less fine-grained.

Other data from studies such as Alexandropoulou et al. (2022) and Cremers (2022) also exist, but they will not be discussed here.

4. Theories and Approaches:

Cremers (2022) provides a clear summary of the different approaches in the literature. He identifies two main approaches: *lexical* and *pragmatic*. In the lexical approach, the properties of the scale determine whether an adjective is absolute or relative. In the pragmatic approach, the main goal is to derive or calculate a distribution of measurements within the comparison class.

He then discusses a debate regarding the nature of these cognitive processes. Are they simple *heuristics* or sophisticated *rational reasoning*? According to Cremers, pragmatic approach depends on two theories. One assumes rationality at the speaker's level. The second theory, known as evolutionary theory, focuses on long-term optimality. The first theory aligns with the RSA model by Lassiter and Goodman (2013), while evolutionary theories correspond to the SOM model by Qing and Franke (2014). He continues by stating that lexical approaches tend to rely on heuristics.

Since this project will use the RSA model without incorporating evolutionary theory, it aligns with the first category of the pragmatic approach, which assumes rationality at the speaker's level.

B. Plans for the Project :

1. Gap in the Literature :

While much of the existing work, such as that by Lassiter and Goodman (2013) and Qing and Franke (2014), focuses on the contextual variability of thresholds and the relationship between scale structure and the absolute/relative distinction, few studies have specifically addressed the role of contextual comparison classes in a computationally tractable framework, especially with the exclusion of intensifiers or negation.

While RSA has been applied to model vagueness and context sensitivity in various contexts, including in the works of Lassiter and Goodman (2013), Zhao (2019), and Xiang et al. (2022), the literature often assumes the presence of intensifiers or other elements that modify the meaning of adjectives. However, these elements (such as "very" or "not") are not the focus of the current study. The gap, therefore, lies in the need for a model that can isolate the effect of context-dependent thresholds in gradable adjectives, focusing specifically on their positive forms without the complicating factors of intensifiers or negation.

Additionally, while previous work has examined the relationship between thresholds and comparison classes, few studies have explicitly compared class posteriors with threshold posteriors to explore how these two concepts interact and contribute to vagueness and borderline cases. This project will fill this gap by focusing on the positive form of adjectives and comparing these posteriors across different comparison classes, contributing new insights into the dynamics of vagueness and context sensitivity in gradable adjectives.

2. Data :

This project will use synthetic data, generated by a computer, similar to the data from Leffel et al. (2019). However, there will be significant differences. For instance, they used both negation and intensifiers to study vagueness and conversational implicatures. This project will examine vagueness and context sensitivity without negation and intensifiers. Instead, it will focus on the positive form. They used both relative and absolute adjectives. This project will also use both.

The synthetic sentences will closely resemble those used in the study by Leffel et al. (2019). Examples of these sentences include 'John is tall' and the exact height of John, e.g., 1.80. The rest of the project will differ. After generating the synthetic data, an RSA model similar to that of Lassiter and Goodman (2013) will be deployed. If feasible, a process similar to that in Zhao (2019) will be implemented. In Zhao's study, participants adjusted a slider on a scale from 0 to 100 to indicate their agreement with sentences like 'Ronald is tall,' based on Ronald's exact height. Alternatively, class posteriors will be compared with threshold posteriors.

3. Model :

The Rational Speech Act (RSA) model, developed by Lassiter and Goodman (2013), is particularly suited for modeling vagueness and context sensitivity in gradable adjectives due to its ability to account for context-dependent reasoning. The RSA model is based on Bayesian inference, allowing speakers and listeners to infer meaning in a probabilistic manner, which is ideal for handling the uncertainties and variabilities inherent in the interpretation of gradable adjectives.

One of the key advantages of RSA is its recursive structure, which models the interaction between the speaker and listener. This recursive reasoning allows the model to handle situations where context plays a critical role in determining the interpretation of an adjective. In the case of gradable adjectives, the threshold for terms like “tall” or “rich” is not fixed but varies depending on the comparison class (e.g., tall among athletes vs. tall among trees). The RSA framework naturally accommodates this context-sensitivity by allowing the listener to infer the most likely interpretation based on the speaker's intended meaning, considering both the context and the speaker's prior knowledge.

Furthermore, RSA allows for the prediction of contextual shifts in threshold values, which is central to understanding the vagueness of gradable adjectives. The model's probabilistic nature enables the calculation of threshold posteriors, which can be compared across different contexts, helping to answer key questions about how thresholds change with different comparison classes. The flexibility of the RSA model in handling such dynamic, context-dependent thresholds makes it an ideal tool for investigating the vagueness and borderline cases that arise in the interpretation of gradable adjectives.

In this model, the main goal is to derive the threshold variable as a probability distribution using Bayesian Inference. Next, class posteriors and threshold posteriors will be calculated to examine vagueness and borderline cases in gradable adjectives. Specifically, when the comparison class for a gradable adjective changes, a second class posterior and a second threshold posterior will be calculated. These two calculation results will then be plotted. By comparing the two plots, we will be able to answer the following questions:

1. Is the gradable adjective vague?
2. Are there any borderline cases?
3. Are the thresholds context-dependent?
4. Does the scale structure determine the relative/absolute adjective distinction?

References :

- Lassiter, D., & Goodman, N. D. Context, scale structure, and statistics in the interpretation of positive-form adjectives. *Proceedings of SALT*, 23(0), (2013). <https://doi.org/10.3765/salt.v23i0.2658>
- Leffel, Timothy & Cremers, Alexandre & Gotzner, Nicole & Romoli, Jacopo. Vagueness in Implicature: The Case of Modified Adjectives. *Journal of Semantics* 36, (2019). [10.1093/jos/ffv020](https://doi.org/10.1093/jos/ffv020)
- Kennedy, C. Vagueness and grammar: the semantics of relative and absolute gradable adjectives. *Linguistics & Philosophy* 30, 1–45, (2007). <https://doi.org/10.1007/s10988-006-9008-0>
- Burnett, H. A Delineation solution to the puzzles of absolute adjectives. *Linguist and Philos* 37, 1–39 (2014). <https://doi.org/10.1007/s10988-014-9145-9>
- Qing, C., & Franke, M. Gradable adjectives, vagueness, and optimal language use: A speaker-oriented model. *Proceedings of SALT* 24, 23–41, (2014). <https://doi.org/10.3765/salt.v24i0.2412>
- Zhao, Zhuoye. Interpreting Intensifiers for Relative Adjectives: Comparing Models and Theories. (2019). [10.1007/978-3-662-59620-3_14](https://doi.org/10.1007/978-3-662-59620-3_14)
- Xiang, M., Kennedy, C., Xu, W., & Leffel, T. Pragmatic reasoning and semantic convention: A case study on gradable adjectives. *Semantics and Pragmatics* 15, 1–64 (2022). <https://doi.org/10.3765/sp.15.9>
- Cremers, A. Interpreting gradable adjectives: Rational reasoning or simple heuristics? *Empirical Issues in Syntax and Semantics* 14, to appear (2022). <https://semanticsarchive.net/Archive/mZlMWMwY/Cremers%20-%20EISS%20-%20Alexandre%20Cremers.pdf>